

Question ID: 997bec28

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

The perimeter of an isosceles triangle is **83** inches. Each of the two congruent sides of the triangle has a length of **24** inches. What is the length, in inches, of the third side?

Correct Answer: 35

Rationale

The correct answer is **35**. It's given that the perimeter of an isosceles triangle is **83** inches and that each of the two congruent sides has a length of **24** inches. The perimeter of a triangle is the sum of the lengths of its three sides. The equation **$24 + 24 + x = 83$** can be used to represent this situation, where **x** is the length, in inches, of the third side. Combining like terms on the left-hand side of this equation yields **$48 + x = 83$** . Subtracting **48** from both sides of this equation yields **$x = 35$** . Therefore, the length, in inches, of the third side is **35**.